

# Georgia Tech Academic AI Playbook

**2026 Edition**



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**Title:** AI-Powered Analysis of Student Feedback for Higher Education Decision-Making

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## Formation

Student surveys and personalized responses form a critical foundation for both institutional improvement and student success. Surveys generate the structured and unstructured data necessary to guide decision-making at scale, while personalized responses help translate that data into a more relational and pedagogical tool that supports deeper learning. In this sense, surveys act as a bridge between institutional assumptions and the lived realities of students, making them a cornerstone of modern higher education practice (Shah et al., 2017).

They are required for several critical institutional functions such as:

- **Accreditation and Quality Assurance:** Accrediting bodies require institutions to use systematic data to document improvements over time. Tools like the Student Satisfaction Inventory (SSI) provide the objective evidence needed for these reports, ensuring that institutions are identifying and addressing their challenges (Ruffalo Noel Levitz, 2021).
- **Informing Pedagogical Change:** Surveys allow instructors to identify specific strengths and weaknesses in their curriculum and teaching methods. This data enables data-informed course improvements rather than relying on anecdotal evidence (Jensen & Roof, 2016).
- **Institutional Assessment and Marketing:** Universities use survey results to promote their strengths in ranking tables and to guide decisions regarding internal funding and faculty rewards for teaching excellence.
- **Continuous Improvement Loops:** Surveys are essential for Action Research, the cyclical process where educators reflect, plan, and act to solve local classroom problems (Johnson & Christensen, 2024).

Despite their centrality, institutions continue to struggle with effectively operationalizing survey data, particularly as course sizes and program scales increase. The growing volume of qualitative feedback makes it increasingly difficult for instructors and administrators to systematically review responses, identify meaningful patterns, and translate insights into timely instructional action (Watt, 2007). As a result, much of this feedback is still processed manually or semi-manually, with themes typically synthesized only after a course has concluded, limiting the ability to respond to student needs in real time. It is within this gap between data collection and actionable interpretation that this initiative is situated. This challenge has become even more pronounced with the widespread adoption of large-enrollment, hybrid, and online learning environments. These instructional models rely heavily on digital platforms that continuously generate substantial volumes of qualitative data through course evaluations, mid-semester surveys, reflection assignments, discussion forums, and program reviews. While these data

sources are rich in insight and capture dimensions of student experience that quantitative metrics often miss, they also amplify the challenge of making sense of feedback at scale and in real time.

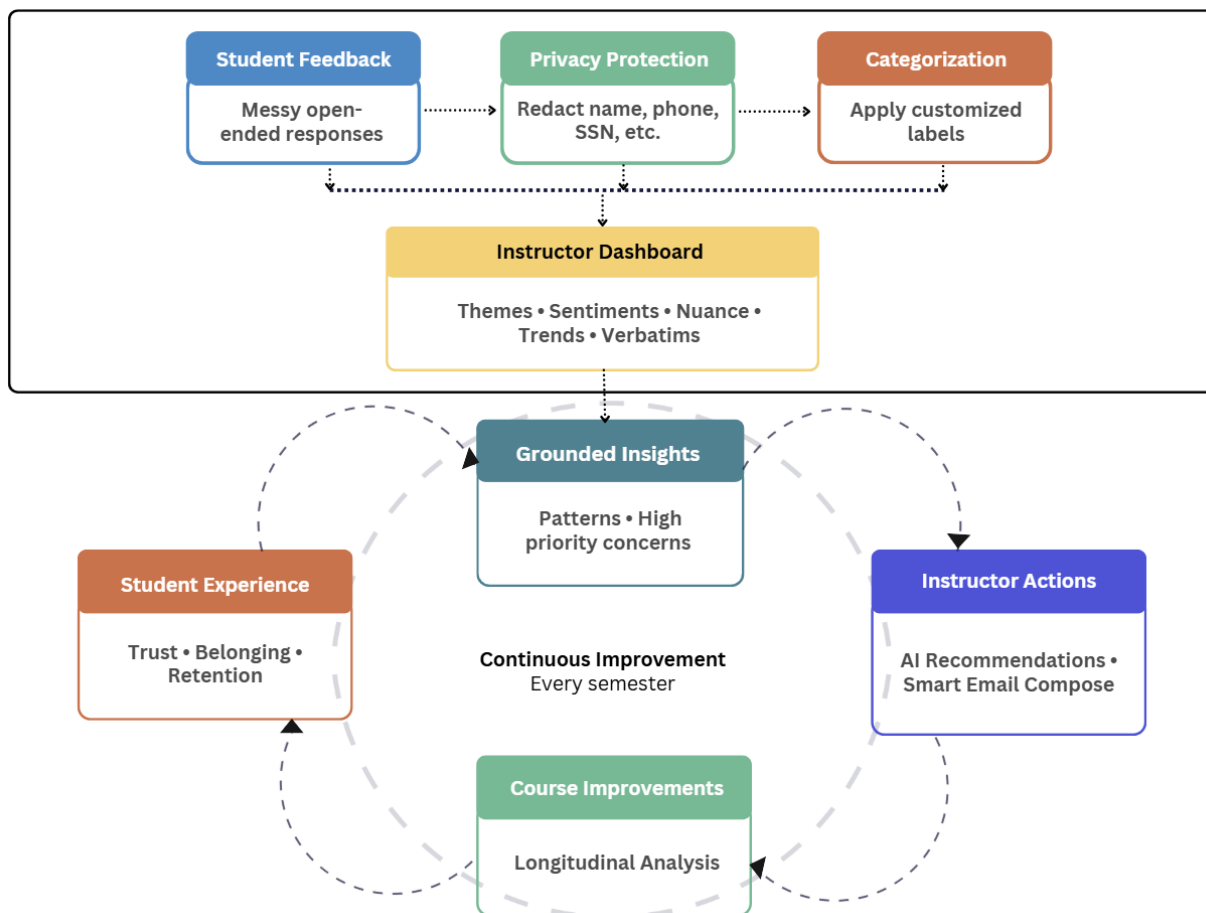
The initiative sits at the intersection of educational technology, learning analytics, and natural language processing within the broader field of learning engineering. It responds to recent advances in transformer-based models and large language models that enable more nuanced analysis of unstructured text, alongside a growing institutional emphasis on continuous improvement and evidence-based teaching. Together, these shifts create an opportunity to bridge the gap between collecting student feedback and transforming it into timely, actionable insights that can directly inform teaching practice.

## **Action**

This initiative resulted in the development of an AI-powered platform that helps higher education institutions analyze and respond to large volumes of open-ended student feedback. The product's current direction has been informed by customer discovery interviews with multiple stakeholder groups across higher education. Development has been supported through several grants and institutional programs, including Georgia Tech's Quadrant-I and university innovation centers, which have provided valuable feedback and opportunities for early pilot adoption.

The platform consists of three integrated components aligned with common institutional workflows. First, an automated categorization engine processes student responses and classifies them into both predefined and emerging themes within a higher education-focused taxonomy, enabling structured interpretation of qualitative feedback at scale. Second, an instructor-facing analytics dashboard organizes categorized feedback into an interpretable format, allowing instructional teams to identify recurring themes, monitor trends over time, and review anonymized responses. Analysis is generated in near real time, with reports typically available within two to three minutes of data upload, enabling instructors to quickly review and download actionable insights. Third, an AI-assisted communication and recommendation tool helps close the feedback loop by generating draft outreach messages and suggesting potential actions based on institutional resources and instructor-provided materials. These recommendations support timely, context-aware responses to student concerns while maintaining full human oversight of all communications and decisions.

As illustrated in Figure 1, the platform is organized into two complementary zones. The upper zone represents the technical processing pipeline, where student feedback is automatically prepared, anonymized, categorized, and transformed into actionable insights. These activities occur behind the scenes and require no technical expertise from instructors. The lower zone represents the pedagogical improvement cycle, where educators review insights, make instructional adjustments, and observe resulting changes in student learning and course experience. Together, these zones reflect a core design principle of the platform: automating technical complexity so educators can focus on teaching and decision-making.



*Figure 1: Design Flow Diagram for Feedback Fusion*

The platform was developed and tested using authentic course survey data from existing institutional feedback systems, allowing evaluation within real instructional workflows. Development combines commercial AI tools, open-source machine learning libraries, and modern web technologies. The system is currently being migrated to AWS-native infrastructure to support scalability, security, and institutional compliance requirements. In addition, several proprietary models and workflows were developed to better capture nuance in student feedback, including custom categorization approaches, taxonomies, and feedback structuring methods tailored specifically for higher education contexts.

No formal changes were made to course syllabi, grading policies, or existing instructional structures as part of this initiative. The platform was designed to complement existing teaching practices by improving instructors' ability to interpret and respond to student feedback within their current workflows. Rather than replacing existing feedback collection processes, the system allows instructors to upload, or process exported feedback data from current institutional systems and receive structured insights without requiring changes to how feedback is collected. This approach reduces disruption to established course processes while enabling instructors to incorporate data-informed insights into ongoing instructional decision-making.

Product development was informed through conducting roughly 70 structured online interviews with instructors, teaching assistants, institutional leaders, students, and qualitative research practitioners.

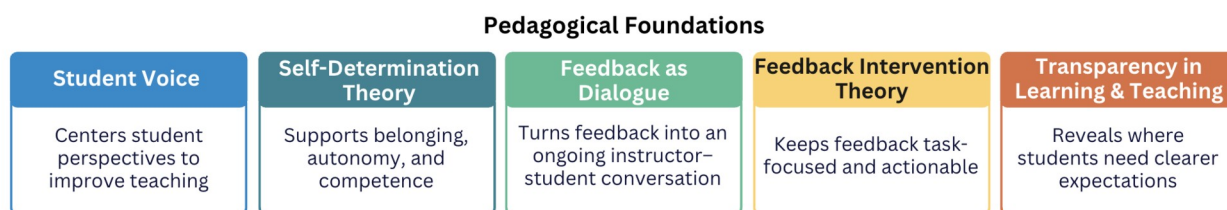


Interviews were conducted with both members of the Georgia Tech community as well as individuals at other institutions. Interview transcripts were thematically coded across groups to uncover insights into the problem and proposed solution.

## Result

The Feedback Fusion platform has been successfully validated at the minimum viable product (MVP) stage and is currently undergoing further development, demonstrating strong alignment with instructional needs in higher education settings. Evaluation using authentic course survey data showed that the system can reliably organize and structure large volumes of open-ended student feedback in ways that closely align with instructor-generated interpretations and categorizations.

Early testing indicated that faculty spent less time reviewing qualitative responses, identified recurring student concerns more quickly, and gained earlier awareness of students who may benefit from academic or personal support. These capabilities enabled timelier, in-course responses to student feedback rather than relying solely on post-course reflection, improving instructors' ability to address concerns while learning was still taking place. These outcomes align with established research on teaching and learning, as illustrated in Figure 2, including student voice, which centers student perspectives in course decisions (Mitra, 2018); self-determination theory, which supports autonomy, competence, and relatedness (Deci & Ryan 2012; Lan & Hew, 2020); feedback as dialogue, which treats feedback as an ongoing exchange rather than one-way transmission (Carless, 2016); feedback intervention theory, which shows that timely, task-focused feedback improves learning (King, 2015; Kluger & DeNisi, 1996); and transparency in learning and teaching, where clear expectations build student trust (Winkelmes, 2023).



*Figure 2: Alignment of Feedback Fusion with Key Educational Research Concepts*

Our customer discovery interviews with instructors, TAs, student support staff, and academic leaders consistently highlighted a significant unmet need for scalable qualitative feedback analysis. Participants across several institutions indicated that while open-ended student feedback is highly valuable, manually reviewing large volumes of comments is time-consuming and often impractical. Stakeholders emphasized that AI solutions must support, rather than replace, human interpretation by providing transparent outputs and supporting evidence that preserve user control. Interviews also revealed that seamless workflow integration with existing LMS platforms is critical for adoption, as disconnected or manually intensive tools are unlikely to be used consistently. Additionally, respondents identified student participation barriers as a challenge for collecting meaningful feedback, suggesting that more accessible channels and participation incentives could improve response rates. These findings point to a clear market opportunity for a privacy-conscious, LMS-integrated platform that combines scalable qualitative analysis with transparent, human-centered workflows.

At the time of writing, the platform has not yet been deployed at scale through institutional systems such as Canvas. Broader deployment is planned as development and integration efforts continue. Expanded

deployment will also create opportunities for systematic evaluation of the platform's impact. Planned measures include the number of students and courses supported, response turnaround times, instructor engagement with feedback, and changes in teaching practices and student learning experiences over time. This data will help inform ongoing product refinement and provide evidence of the platform's effectiveness in supporting feedback-informed instruction.

## Lessons

- **Lesson 1: Start by listening to your users.** This initiative revealed the importance of conducting extensive customer discovery before developing a solution. These interviews revealed a strong demand for tools that help educators analyze large volumes of open-ended student feedback while maintaining human oversight and transparency. Participants emphasized the importance of seamless LMS integration, actionable insights, and privacy-conscious workflows to support adoption. These findings validate the need for a scalable, human-centered approach to qualitative feedback analysis in higher education.
- **Lesson 2: Different stakeholders bring different perspectives.** Engaging a wide range of stakeholders revealed that different groups often have distinct needs and expectations. Our conversations helped challenge assumptions and uncover nuances that may have otherwise been overlooked. This process highlighted the importance of gathering perspectives from multiple stakeholder groups, narrowing the project scope, and iteratively refining ideas based on feedback to make more informed and objective decisions in an inherently subjective domain.
- **Lesson 3: AI works best as a partner and not as a replacement:** Our conversations consistently highlighted that AI-generated insights are most valuable when they augment, rather than replace, instructor expertise and judgment. Participants emphasized the importance of transparency, interpretability, and human oversight in educational applications of AI. These findings reinforced the need for human-in-the-loop workflows that allow instructors to review, validate, and act on generated insights while maintaining control over instructional decisions.

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